TEXORA · LECTURE SHEET

Variance Analysis and Financial Statement Analysis

Closing the control loop: material, labour and overhead variances computed and reconciled, then the reading of financial statements — horizontal and vertical analysis and a full set of ratios worked from one balance sheet.

AC-4205 · COST AND MANAGEMENT ACCOUNTING · TOPIC 9 · LEVEL 4 TERM 2

1. Variance analysis: the idea

A **variance** is the difference between a standard (Topic 5) and the actual result for the same output: **favourable (F)** when actual cost is below standard or revenue above, **adverse (A)** otherwise. Variance analysis splits each total difference into a **price/rate** part (paid more or less than standard) and a **usage/efficiency** part (used more or less than standard) — because different managers answer for each, and the remedies differ entirely.

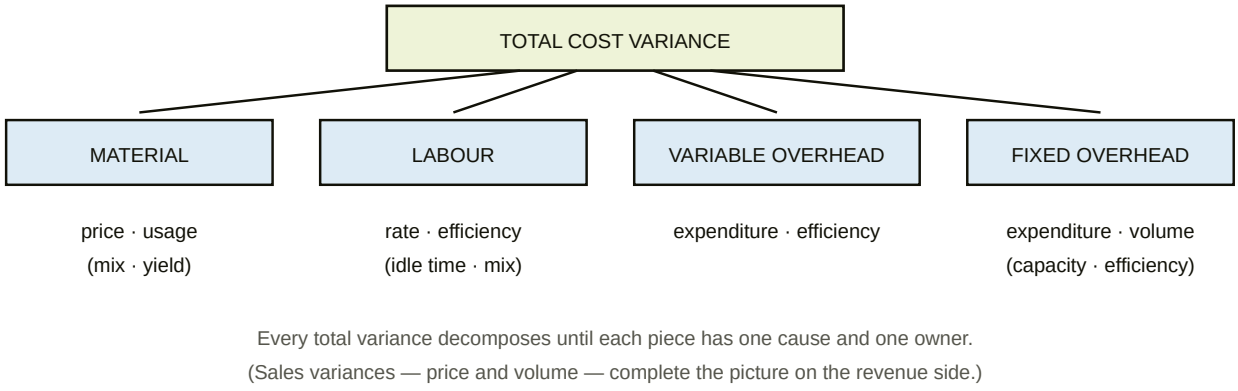


Figure 1 — The variance tree. Investigation follows the branches: a price variance goes to purchasing, a usage variance to production, a rate variance to wage negotiation, an efficiency variance to the line.

2. Material and labour variances — worked example

$$\text{Material price variance} = (\text{Standard price} - \text{Actual price}) \times \text{Actual quantity}$$

$$\text{Material usage variance} = (\text{Standard quantity for actual output} - \text{Actual quantity}) \times \text{Standard price}$$

$$\text{Labour rate variance} = (\text{Standard rate} - \text{Actual rate}) \times \text{Actual hours};\quad \text{Labour efficiency variance} = (\text{Standard hours} - \text{Actual hours}) \times \text{Standard rate}$$

WORKED EXAMPLE — MATERIAL AND LABOUR VARIANCES FOR A BATCH

Output: 1,000 units. Material standard: 5 kg per unit @ Tk 200 → standard quantity 5,000 kg, standard cost Tk 10,00,000.

Actual: 5,400 kg used, bought at Tk 190 → actual cost Tk 10,26,000.

Price variance = (200 − 190) × 5,400 = **Tk 54,000 F** (bought cheaper than standard).

Usage variance = (5,000 − 5,400) × 200 = **Tk 80,000 A** (used 400 kg too much).

Net material variance = 54,000 F − 80,000 A = **Tk 26,000 A**. Check: 10,26,000 − 10,00,000 = Tk 26,000 A ✓.

Story the split tells: purchasing saved money — possibly by buying a lower grade — while production consumed 8 % excess material. The two facts belong to different desks, and one may have caused the other.

Labour standard: 4 hours per unit @ Tk 150 → standard hours 4,000, standard cost Tk 6,00,000.

Actual: 4,300 hours @ Tk 145 → Tk 6,23,500.

Rate variance = (150 − 145) × 4,300 = **Tk 21,500 F**.

Efficiency variance = (4,000 − 4,300) × 150 = **Tk 45,000 A**.

Net labour variance = **Tk 23,500 A**. Check: 6,23,500 − 6,00,000 = Tk 23,500 A ✓.

Cheaper labour (rate F) that works slower (efficiency A) is a classic false economy — only the split reveals it.

Overhead variances follow the same grammar: variable overhead splits into expenditure (spent more/less) and efficiency (driven by the absorption base, usually labour hours); fixed overhead splits into expenditure (budget vs actual spend — e.g. budget Tk 5,00,000 against actual Tk 5,40,000 is Tk 40,000 A) and volume (output above/below the budgeted level at which the rate was set). **Investigation rules** keep the system sane: investigate by size and controllability, remember variances interact (cheap material ↔ excess usage), and treat variance reports as questions, not verdicts.

3. Analysis of financial statements

Horizontal analysis compares each line across periods (growth, trend); **vertical (common-size) analysis** expresses each line as a percentage of a base within one period (sales in the income statement, total assets in the balance sheet) so structures can be compared across firms and years. **Ratio analysis** then relates pairs of figures into the standard families:

WORKED EXAMPLE — RATIOS OF A TEXTILE COMPANY

Sales Tk 60,00,000 · Cost of goods sold Tk 42,00,000 · Profit before interest & tax Tk 6,00,000 · Net profit Tk 4,80,000.
Current assets Tk 15,00,000 (inventory Tk 7,00,000, receivables Tk 6,00,000, cash Tk 2,00,000) · Current liabilities Tk 7,50,000 · Fixed assets Tk 30,00,000.
Gross profit ratio = (60,00,000 − 42,00,000) ÷ 60,00,000 = **30 %**.
Net profit ratio = 4,80,000 ÷ 60,00,000 = **8 %**.
Current ratio = 15,00,000 ÷ 7,50,000 = **2.0 : 1**.
Quick ratio = (15,00,000 − 7,00,000) ÷ 7,50,000 = **1.07 : 1** — liquid cover survives even if stock cannot be sold.
Inventory turnover = 42,00,000 ÷ 7,00,000 = **6 times** → stock held ≈ 365 ÷ 6 = **61 days**.
Receivables collection = 6,00,000 ÷ 60,00,000 × 365 = **36.5 days** of sales outstanding.
Capital employed = (15,00,000 + 30,00,000) − 7,50,000 = Tk 37,50,000; **ROCE** = 6,00,000 ÷ 37,50,000 = **16 %**.
Read together: healthy liquidity (2.0 / 1.07), stock turning every two months, customers paying in about five weeks, and 16 % earned on the capital tied up — each ratio a question (is 61 days of stock right for this order book?) rather than a grade.

FAMILY	KEY RATIOS	QUESTION ANSWERED
Profitability	GP %, NP %, ROCE, return on equity	Does the business earn enough on sales and on capital?
Liquidity	Current, quick	Can short-term bills be paid as they fall due?
Efficiency	Inventory turnover/days, receivables days, payables days, asset turnover	How hard do the assets and working capital work?
Solvency / gearing	Debt–equity, interest cover	Can the capital structure carry its obligations?

Limitations close the course honestly: ratios inherit the accounting policies and window-dressing of the statements beneath them, a single ratio proves nothing without trend and industry comparison, and inflation and seasonality distort point-in-time figures. Analysis points to where to dig — the digging is operational, in the cost records of Topics 1–8.

Glossary

Variance — standard minus actual (favourable or adverse). **Price/rate variance** — the rate effect at actual quantity/hours. **Usage/efficiency variance** — the quantity effect at standard rate. **Horizontal analysis** — comparison across periods. **Vertical (common-size) analysis** — lines as percentages of a

base within one period. **ROCE** — return on capital employed. **Quick ratio** — current assets excluding inventory, over current liabilities. **Inventory turnover** — COGS ÷ average (or closing) inventory.

Key points

• Every variance splits into price/rate and usage/efficiency: material Tk 54,000 F price against Tk 80,000 A usage (net Tk 26,000 A); labour Tk 21,500 F rate against Tk 45,000 A efficiency (net Tk 23,500 A) — both nets reconcile exactly to actual-minus-standard cost.
• Splitting assigns each difference to its cause and its owner; totals alone blame everyone and therefore no one.
• Statement analysis: horizontal for trend, vertical for structure, ratios for relationships — the worked firm shows GP 30 %, NP 8 %, current 2.0, quick 1.07, stock 61 days, collection 36.5 days, ROCE 16 %.

Practice

1. Standard: 2 kg per unit @ Tk 150; output 1,000 units; actual usage 2,150 kg bought at Tk 142. Compute price, usage and net material variances and verify the net against actual and standard total costs.
2. From: sales Tk 25,00,000, COGS Tk 17,50,000, current assets Tk 6,00,000, current liabilities Tk 4,00,000 — compute the gross profit ratio and the current ratio, with one interpretive sentence each.
3. A labour rate variance is favourable and the efficiency variance adverse by more. Give two operational explanations, and say what you would check before blaming the workforce.
4. State two limitations of ratio analysis and how an analyst mitigates each.

Answer key

1. Standard quantity = 2,000 kg; standard cost = 2,000 × 150 = Tk 3,00,000. Actual cost = 2,150 × 142 = Tk 3,05,300.
   Price variance = (150 − 142) × 2,150 = **Tk 17,200 F**. Usage variance = (2,000 − 2,150) × 150 = **Tk 22,500 A**. Net = **Tk 5,300 A**. Verification: 3,05,300 − 3,00,000 = Tk 5,300 A ✓.
2. GP ratio = (25,00,000 − 17,50,000) ÷ 25,00,000 = **30 %** — thirty poisha of every sales taka survives direct production cost to meet overheads and profit. Current ratio = 6,00,000 ÷ 4,00,000 = **1.5 : 1** — short-term assets cover short-term liabilities one and a half times; adequate on its face, to be read with the quick ratio and the stock position before judging safety.
3. Explanations (any two): a lower-paid, less-skilled crew was engaged (cheaper rate, slower work); overtime was replaced by additional basic-rate hours from inexperienced hires; wage rates fell market-wide while machine condition or material quality slowed the line — the inefficiency may not

be labour's at all. Checks before blaming: machine downtime and maintenance records, material quality/usage variances for the same period, whether standards were revised for rate changes, and idle-time records — efficiency variance measures hours, not effort.

4. Any two, with mitigations: (1) Ratios inherit accounting policies/window dressing — compare with cash-flow data and read the notes and policies. (2) A point-in-time ratio is distorted by seasonality — use average balances and compare like dates across years. (3) No universal benchmarks — compare against industry and the firm's own trend. (4) Inflation mixes values of different years — focus on relationships and physical measures alongside money ratios.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib